

Kasra Fallah

Ph.D. Candidate, Electrical Engineering
Columbia University, New York, USA

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Objective

Ph.D. candidate in Electrical Engineering developing a research program at the intersection of control theory, machine learning, and stochastic systems. My long-term goal is to understand how learning, feedback, and structural knowledge can be combined to build intelligent systems that adapt and make reliable decisions in uncertain and dynamically evolving environments. I am particularly interested in developing fundamental theory and algorithms that bridge learning and control, while using these principles to study complex systems ranging from autonomous and multi-agent systems to neural computation.

Education

Columbia University, New York, USA 2024 – Present
Ph.D. in Electrical Engineering (GPA: 4.05/4.0) Advisor: Prof. James Anderson

Passed the Doctoral Qualifying Examination with high distinction in Signals, Systems, and Communications.

University of Waterloo, Waterloo, Canada 2022 – 2024
M.A.Sc. in Electrical Engineering (GPA: 4.0/4.0) Advisor: Prof. Ravi Mazumdar

Thesis: *On the Calculation of Mutual Information for Channels with Gauss–Markov Noise.*

Sharif University of Technology, Tehran, Iran 2018 – 2022
B.Sc. in Electrical Engineering (GPA: 3.95/4.0, Rank: 13)

Selected Publications

- **K. Fallah**, L. F. Toso, J. Anderson, “On the Gradient Domination of the LQG Problem,” *Under review*, 2025.
- **K. Fallah***, L. F. Toso*, C. Stamouli*, G. J. Pappas, J. Anderson, “Multitask LQG Control: Performance and Generalization Bounds,” *IEEE Conference on Decision and Control (CDC)*, 2026.
- **K. Fallah***, L. F. Toso*, J. Anderson, “Adversarially Robust Multitask Adaptive Control,” *Learning for Dynamics and Control (L4DC)*, 2026.
- **K. Fallah***, H. N. Chen, R. Singha, E. Kong, G. Turi, A. Losonczy, E. Zabeh*, “NeuralField-Manifold: Reconstruction of LFP Manifold with Lag Embedding,” *NeurIPS*, 2026.

- M. Parson-Scherban, **K. Fallah**, N. Rahbariasr, B. Steyaert, J. Anderson, M. Preindl, “On the Unification of Optimal Current Reference Theory for Wound Rotor Synchronous Machines,” *IEEE Conference on Decision and Control (CDC)*, 2026.
- M. Parson-Scherban, **K. Fallah**, N. Rahbariasr, P. Wang, B. Steyaert, J. Anderson, M. Preindl, “Applying Linear Parameter-Varying Model Predictive Current Control to an Enhanced WRSM Model,” *IEEE/CCAE Symposium on Advanced Electrification Systems (SAES)*, 2026.
- **K. Fallah**, “On the Calculation of Mutual Information for Channels with Gauss–Markov Noise,” *M.A.Sc. Thesis, University of Waterloo*, 2024.

Awards and Honors

- PhD Graduate Studies Fellowship, Columbia university, 2024
- Xuemin Shen Graduate Scholarship in Telecommunication, 2023
- International Master’s Award of Excellence (CAD 7500), 2022 & 2023
- Graduate Research Scholarship (CAD 64,000)
- Gold Medalist, 30th National Physics Olympiad (2017)
- Gold Medalist, 15th National Rubocup competition(2015)

Research Experience

Ph.D. Researcher

2024 – Present

Columbia University

Advisor: Prof. James Anderson

Developing theoretical foundations for learning and optimization in stochastic and partially observed dynamical systems, including global convergence guarantees for policy optimization in Linear Quadratic Gaussian (LQG) control.

Studying multitask, adaptive, and robust control under system heterogeneity, with an emphasis on performance, generalization, and regret guarantees for learning shared policies across related dynamical systems.

Research Fellow

Summer 2026

UT Southwestern Medical Center

Mentors: Prof. Wenhao Zhang and Dr. Erfan Zabeih

Developing geometric and topological methods for recovering latent structure in neural population dynamics, including lag-embedding approaches for reconstructing low-dimensional manifolds directly from local field potential (LFP) recordings.

Developing control-theoretic models of neural sampling and recurrent computation, using closed-loop inhibitory feedback and oscillatory modulation to study and accelerate convergence of neural inference dynamics.

Research Assistant

2022 – 2024

University of Waterloo

Supervisor: Prof. Ravi Mazumdar

Developed information-theoretic and stochastic-filtering methods for continuous-time communication channels with correlated Gaussian noise, connecting mutual information to estimation error through state-space and filtering representations.

Research Intern

Jul 2021 – Sep 2022

MCI Lab (Mobile Telecommunication Co. of Iran)

Supervisor: Dr. M. Fakharzadeh

Developed software-defined radio simulations and experimental pipelines using bladeRF, MATLAB/Simulink, and GNU Radio for wireless-channel modeling and mmWave communication systems.

Analyzed and tested transmitter–receiver architectures for 30,GHz commercial 5G communication systems.

RD Intern

May 2021 – Sep 2021

Telecommunication Company of Iran (TCI)

Developed simulation-based methods in Python and MATLAB for analyzing network switching and communication-traffic optimization.

Teaching Experience

Head Teaching Assistant

2024 – 2025

Columbia University

Introduction to Control Theory; Modern Control Theory

Head Teaching Assistant

2023

University of Waterloo

Probability Theory and Statistics II

Head Teaching Assistant

2019 – 2022

Sharif University of Technology *Signals and Systems; Electric Circuits; Numerical Methods; Engineering Mathematics*

Physics Olympiad Tutor

2017 – 2022

Young Scholars Club and Selected High Schools

Service

- **Leadership:** Electrical Engineering Ambassador, Columbia University, 2026.
- **Reviewer (Control Community):** Conferences: IEEE Conference on Decision and Control (CDC), American Control Conference (ACC), Learning for Dynamics & Control (L4DC); Journals: IEEE Transactions on Information Theory (TIT), IEEE Transactions on Automatic Control (TAC), IEEE Control Systems Letters (L-CSS).
- **Reviewer (CS/ML Community):** Conferences: NeurIPS/ ICML

Languages

- English (full professional), Persian/Farsi (native), Gilaki (bilingual).

Technical Skills

Programming: Python, MATLAB/Simulink, C/C++

Scientific Computing & ML: PyTorch, JAX, NumPy/SciPy, CVXPY, GPU computing

Tools: Git, Linux, $\text{\LaTeX}$